

VL-DINO: Leveraging CLIP Vision-Language Knowledge for Open-Vocabulary Object Detection

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Abstract

Vision-language models like CLIP can provide rich semantic priors for open-vocabulary object detection. However, jointly integrating both textual and visual knowledge into detection architectures remains challenging. In this paper, we propose **VL-DINO**, an open-vocabulary detector that enhances DINO through more effective exploitation of CLIP’s *Vision-Language* knowledge. Specifically, a *Query-guided Positive Sample Construction* (QPSC) module is first developed to construct additional high-quality positive samples, enabling the vanilla DINO framework to better accommodate mixed training across heterogeneous data sources while providing more vision-language alignment signals, thereby incorporating richer textual knowledge during training. A *Visual Semantic Encoder* (VSE) module is then introduced to distill CLIP visual knowledge into backbone-extracted features, producing fused features for subsequent encoder refinement. Based on the fused features, an *Object-Region Semantic Alignment* (ORSA) module extracts object-centric region features and aligns them with the corresponding textual embeddings, further incorporating textual cues. In the zero-shot setting, VL-DINO-T and VL-DINO-L achieve 36.3 and 38.1 AP on LVIS benchmark, respectively, consistently outperforming prior advanced approaches. Extensive experiments demonstrate the effectiveness and competitive performance of the proposed design. The code and model are available at <https://github.com/HaoZ416/VL-DINO>.

1 Introduction

Open-vocabulary detection(OVD) aims to overcome the inherent limitation of conventional object detectors, whose predictions at inference time are restricted to a predefined set of categories [21, 7, 27]. Recent OVD approaches endow detectors with open-vocabulary generalization by incorporating the textual semantic knowledge of vision-language models(VLM) [26, 13] into traditional detection frameworks to meet the demands of real-world applications [3, 34, 15, 22, 19]. Beyond transferring textual semantic knowledge, existing methods have also explored how to leverage the visual semantic knowledge encoded in the CLIP [26] image encoder. These efforts mainly leverage the image encoder as a teacher to guide the detector, distilling its rich visual knowledge into refined region-level features through feature alignment or supervision signals [38, 8, 6, 37].

However, existing approaches still suffer from several limitations. First, methods for transferring textual semantic knowledge typically depend on layer-wise cross-modal fusion modules, resulting in increased model complexity [22, 19, 3]. Second, methods for exploiting visual semantic knowledge are fundamentally constrained by the pretraining nature of CLIP: trained with coarse-grained image-text supervision, CLIP mainly learns globally aligned visual representations, and its dense visual

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features are therefore suboptimal [43, 35, 10, 17], which weakens the effectiveness of direct feature distillation into detectors. Finally, the joint integration of textual and visual semantic knowledge into a unified detection framework has not been fully explored.

To overcome the above limitations, we propose VL-DINO, an open-vocabulary detector built upon DINO [40] and CLIP. A key obstacle in adapting DINO to OVD lies in its original denoising design, which requires the total number of unique dataset-specific categories to be known in advance. This assumption does not hold in the OVD setting, where grounding datasets [25, 12] are often used to train models, containing a large number of free-form phrases (e.g., “a brown building”) that cannot be enumerated with predefined category indices. Consequently, the application of the original denoising mechanism in DINO becomes infeasible, and the model mainly relies on vanilla object queries to align with textual semantics. In light of this, we develop a *Query-guided Positive Sample Construction* module (QPSC), which constructs additional high-quality positive samples without relying on category-specific labels. QPSC introduces no extra learnable parameters, and all positive samples preserve the denoising mechanism advantage of bypassing bipartite matching while directly contributing to classification and box regression, and meanwhile provides richer vision-language alignment signals, allowing the detector to acquire richer textual semantic knowledge.

Additionally, as discussed above, CLIP mainly captures global semantics, making its dense local features less effective for direct region-level distillation. Motivated by this observation, we introduce a *Visual Semantic Encoder* (VSE), which takes the high-level features extracted by the detector backbone (e.g., Swin Transformer [23]) as input and distill CLIP visual semantics into corresponding feature outputs. This enriches the high-level features with stronger global semantic information, while keeping the low-level detail-rich features unchanged. The distilled high-level features are then fused with their corresponding original backbone features, and together with the unchanged low-level features, are fed into the encoder for further refinement of dense local information.

Finally, we introduce an *Object-Region Semantic Alignment* (ORSA) module, which aligns foreground object-region features of the first encoder layer with their corresponding textual semantics. By doing so, ORSA encourages fused features in the shallow encoder layers to be more sensitive to textual semantic cues, making them better adapted to OVD task while further injecting textual semantic knowledge into the detector.

Overall, these modules enable VL-DINO to effectively absorb the large-scale semantic priors of CLIP from both textual and visual perspectives, rather than relying on a single modality alone. Notably, QPSC and ORSA are used only during training and are discarded at inference time. As a result, beyond DINO detector and CLIP model, only VSE module is retained as an additional component, preserving the overall simplicity of the architecture. Extensive experiments in the zero-shot setting on LVIS [9] and COCO [20] benchmarks demonstrate the effectiveness of the proposed design. VL-DINO-T and VL-DINO-L achieve 36.3 and 38.1 AP on LVIS, and 48.5 and 50.0 AP on COCO, respectively. Our contributions are summarized as follows:

- We propose QPSC, which constructs additional high-quality positive samples to provide DINO with richer textual semantic supervision.
- We introduce VSE, which distills CLIP visual semantic knowledge into high-level backbone features and enhance their global semantic representations.
- We present ORSA, which aligns shallow object-region features with textual embeddings to further integrate textual semantic knowledge into the detector.
- Extensive zero-shot experiments on LVIS and COCO verify the effectiveness of VL-DINO and demonstrate its strong generalizability and competitive performance.

2 Related work

DETR-like Models. DETR [1] introduces Transformer [33] into object detection and establishes a fully end-to-end detection framework. Building upon this paradigm, a series of follow-up works have focused on improving its optimization and representation learning. Deformable DETR [45] extends DETR with deformable attention, enabling more effective modeling of multi-scale visual features in the encoder. To alleviate the training instability caused by Hungarian matching, subsequent methods [18, 40] incorporate denoising mechanisms. Meanwhile, although the one-to-one matching scheme in DETR eliminates hand-crafted designs such as non-maximum suppression and anchor

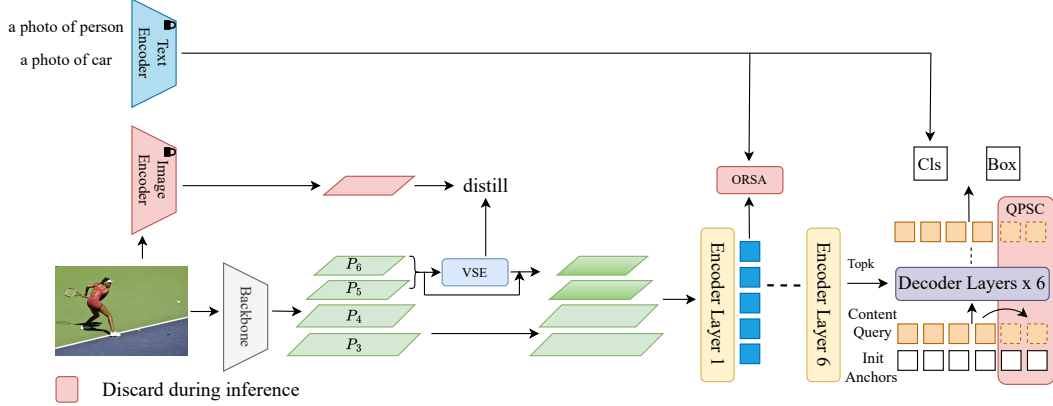


Figure 1: Overview of the VL-DINO model. During training, QPSC leverages reliable content queries to provide additional high-quality positive samples, thereby offering richer textual supervision for vision–language alignment. Given an input image I , the distillation branch employs a CLIP image encoder to extract visual semantic knowledge and distills it into the deep features P_5 and P_6 extracted by the backbone. Meanwhile, ORSA enhances the shallow encoder features by making foreground visual representations more sensitive to textual semantics. During inference, QPSC, the distillation branch, and ORSA are all discarded.

boxes, it also results in sparse supervision. To address this limitation, later works [2, 14, 46, 11] draw inspiration from conventional CNN-based detectors [21, 32] and introduce one-to-many training branches, thereby providing denser supervision signals and improving feature representation.

Open-Vocabulary Detectors. The rise of large-scale pretrained vision-language models [26, 13] has made it practical for OVD detectors to exploit large amounts of vision-language knowledge [15, 42, 44, 39, 28]. A straightforward line of research transfers such semantic knowledge through knowledge distillation [38, 8, 6, 37]. Another representative direction is to unify detection and grounding, allowing the model to benefit from both types of supervision. In this vein, GLIP [19] is among the first to cast detection as grounding and jointly train on detection and grounding datasets, while GroundingDINO [22] further strengthens multimodal understanding through layer-wise cross-modal fusion and decoding. More recently, YOLO-World [3] and YOLOE [34] push OVD toward lightweight and real-time regimes, improving its suitability for deployment on edge devices.

3 Method

In this section, we present the overall framework of VL-DINO, as illustrated in Figure 1, which consists of three core modules. QPSC is first introduced in Section 3.1 to provide high-quality positive samples, followed by VSE in Section 3.2, which injects CLIP-derived visual semantics into backbone-extracted features to produce fused representations. Finally, ORSA in Section 3.3 further refines these representations by aligning object-region features with textual embeddings.

3.1 Query-guided positive sample construction

As discussed in Section 1, the vanilla DINO framework [40] is not well suited for the OVD setting. Inspired by the denoising mechanism [40] and the one-to-many training paradigm [2, 14, 46, 11], we propose QPSC, which constructs high-quality positive samples from reliable content queries to enhance vision-language alignment during training.

To analyze the behavior of object queries, we conduct a preliminary experiment by training DINO model without the denoising mechanism on Objects365v1 dataset [30] until convergence. We then evaluate the model on COCO val2017 split [20] and LVIS minival split [9]. Using the same Hungarian matching strategy as in training, we record, for each query, the total number of matches and its average prediction score. As shown in Figure 2, the matching statistics of object queries show a pronounced long-tailed distribution on both validation sets. Specifically, most matched predictions are concentrated within the first 300 queries, and these queries also yield higher average

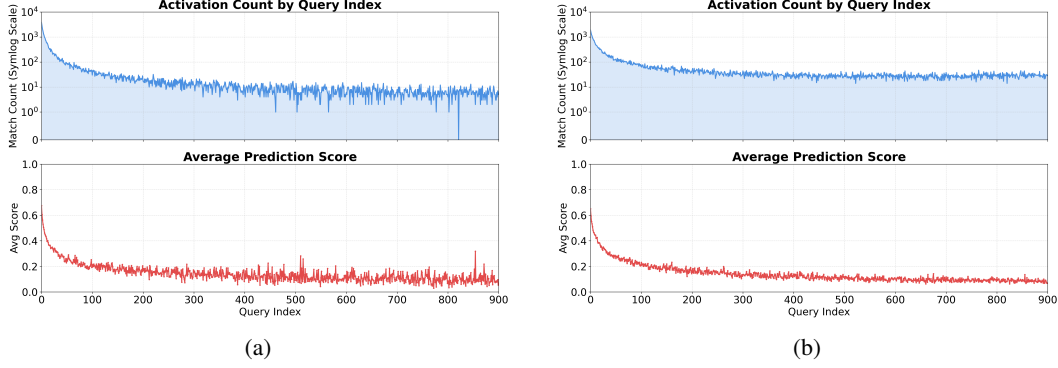


Figure 2: Matching statistics of object queries. The model is trained on Objects365v1 and evaluated on (a) COCO val2017 split and (b) LVIS minival split. In each subfigure, the top plot shows the total matches per query, while the bottom plot shows the average prediction score per query; both exhibit a clear long-tailed distribution.

prediction scores. We hypothesize this behavior stems from the architectural bias of DINO: during training, predictions with higher foreground confidence tend to be implicitly associated with lower-indexed queries, allowing these queries to learn more reliable semantic representations. Based on this observation, the first 300 content queries are selected to construct a reliable query pool.

Following the box denoising in DINO [40], QPSC applies bounded perturbations to each ground-truth box and generates an equal number of auxiliary boxes whose IoU with the corresponding ground truth is greater than 0.5. Each auxiliary box is paired with a content query embedding randomly sampled from the reliable query pool, forming an additional positive object query. These constructed positive object queries are fed into the decoder together with the original object queries and are supervised by both classification and box regression losses.

Notably, each of positive object queries is directly associated with its corresponding ground-truth object, thereby bypassing Hungarian matching during training, and are discarded during inference to avoid any extra computational burden. Furthermore, QPSC focuses exclusively on positive sample construction, while negative supervision remains handled by the original object queries. As a result, model receives richer positive supervision, leading to improved vision-language alignment.

3.2 Visual semantic encoder

We argue that the multi-scale dense features extracted by the Swin Transformer backbone [23] are already highly expressive, and that its two deepest feature maps provide suitable carriers for visual semantic learning. Accordingly, we introduce VSE, composed of three self-attention blocks following the identical architecture of ViT [5] in CLIP [26] and a multi-layer perceptron module. VSE operates on the P_5 and P_6 feature levels to enhance their global semantic capacity. Following CLIP, we prepend a learnable class embedding (CLS) to the input sequence. For each layer $l \in \{5, 6\}$, we input the global class token CLS and the visual feature P_l to obtain

$$\begin{aligned} CLS_l, \bar{P}_l &= \text{SelfAttnBlock}(CLS, P_l), \\ \tilde{P}_l &= P_l + \text{MLP}(\bar{P}_l) \end{aligned} \quad (1)$$

where CLS_l is the output class embedding, $\bar{P}_l$ and $\tilde{P}_l$ are the visually enhanced feature and the fused feature, respectively. In particular, we treat CLS_l as a proxy for distilling CLIP’s visual semantic knowledge into $\tilde{P}_l$. We first optimize a global semantic alignment loss $\mathcal{L}_{\text{sem}}$ to learn semantic similarity between CLS_l and the corresponding CLIP class embedding CLS_{clip} . In addition, we introduce an intra-batch relational loss $\mathcal{L}_{\text{rel}}$ to capture the relative relationships among image samples

within the same batch. The two losses are formulated as follows:

$$\begin{aligned}\mathcal{L}_{sem}^l &= 1 - \frac{1}{B} \sum_{i=1}^B \cos(\|CLS_l^{(i)}\|_2, \|CLS_{clip}^{(i)}\|_2), \\ \mathcal{L}_{rel}^l &= \frac{1}{B} \sum_{i=1}^B KL(r_{clip}^{(i)} \| r_l^{(i)}), \\ r_c^{i,j} &= \frac{\exp(\cos(\|CLS_c^{(i)}\|_2, \|CLS_c^{(j)}\|_2)/\tau_1)}{\sum_{k=1, k \neq i}^B \exp(\cos(\|CLS_c^{(i)}\|_2, \|CLS_c^{(k)}\|_2)/\tau_1)}, \quad j \neq i, c \in \{l, clip\},\end{aligned}\tag{2}$$

Here, i, j , and k are sample indices within the batch, and B is the batch size. τ_1 is a temperature parameter. $KL(\cdot)$, $\cos(\cdot, \cdot)$, and $\|\cdot\|_2$ denote the Kullback–Leibler divergence, the cosine similarity, and the L_2 norm, respectively.

In addition to the aforementioned image-level losses, we further develop a dense spatial semantic loss $\mathcal{L}_{spa}$ to model the spatial semantic relationships between patch tokens and the class embedding:

$$\begin{aligned}\mathcal{L}_{spa}^l &= \frac{1}{B} \sum_{i=1}^B KL(s_{clip} \| s_l), \\ s_c^m &= \frac{\exp(\cos(\|CLS_c\|_2, \|\bar{P}_c^m\|_2)/\tau_2)}{\sum_{n=1}^{H_l \times W_l} \exp(\cos(\|CLS_c\|_2, \|\bar{P}_c^n\|_2)/\tau_2)}, \quad c \in \{l, clip\},\end{aligned}\tag{3}$$

where $H_l \times W_l$ denotes the feature map of spatial size, and m and n index tokens. τ_2 is a temperature parameter. In practice, we apply interpolation to $\bar{P}_{clip}$ to match its spatial resolution with that of $\bar{P}_l$.

The overall distillation loss is defined as

$$\mathcal{L}_{dis} = \frac{1}{2}(\lambda_{sem}(\mathcal{L}_{sem}^6 + \mathcal{L}_{sem}^5) + \lambda_{rel}(\mathcal{L}_{rel}^6 + \mathcal{L}_{rel}^5) + \lambda_{spa}(\mathcal{L}_{spa}^6 + \mathcal{L}_{spa}^5)),\tag{4}$$

where λ_{sem} , λ_{rel} , and λ_{spa} are weighting coefficients that balance the contributions of the three loss terms. The features $\tilde{P}_5$ and $\tilde{P}_6$, enriched with CLIP visual semantic knowledge, serve as global image representations. Together with P_3 and P_4 , they form a new set of multi-scale features, which are fed into the encoder to further refine dense visual representations.

3.3 Object-region semantic alignment

We design ORSA to better adapt the fused feature $\tilde{P}_l$ to detection while further incorporating textual semantic knowledge. Specifically, ORSA consists of a set of learnable queries and a foreground decoder, which follows the same architecture as the decoder but contains only three layers.

During training, the queries are guided by ground-truth box information to extract foreground object features from outputs of the first encoder layer. These features are then aligned with their corresponding textual features. During inference, the ORSA module is removed.

4 Experiments

In this section, we first introduce the implementation details, including the model configuration, training datasets, and evaluation metrics. We then present extensive experimental results and comprehensive analyses to demonstrate the effectiveness of the proposed method and to further understand the contribution of each component.

4.1 Implementation details

Training. VL-DINO is built upon DINO [40] as the detection framework. The overall architecture consists of a Swin Transformer backbone [23], six deformable encoder layers [45], six deformable decoder layers [45], and 900 object queries. For the language and the visual knowledge distillation branch, we use the pre-trained text and vision encoder from CLIP-B/16 [26], whose parameters are

Table 1: Zero-shot object detection performance comparison on LVIS. Cap4M denotes the dataset proposed in GLIP [19], and G-20M denotes the Grounding-20M dataset [29].

Method	Backbone	Training Data	AP/AP ^{Fixed}	AP _r /AP _r ^{Fixed}	AP _c /AP _c ^{Fixed}	AP _f /AP _f ^{Fixed}
GLIP-T [19]	Swin-T	OG	- / 24.9	- / 17.7	- / 19.5	- / 31.0
GLIP-T [19]	Swin-T	OG,Cap4M	- / 26.0	- / 20.8	- / 21.4	- / 31.0
GLIPv2-T [41]	Swin-T	OG,Cap4M	- / 29.0	- / -	- / -	- / -
GDINO-T [22]	Swin-T	OG	- / 25.6	- / 14.4	- / 19.6	- / 32.2
GDINO-T [22]	Swin-T	OG,Cap4M	- / 27.4	- / 18.1	- / 23.3	- / 32.7
G1.5-Edge [29]	EfficientViT-L1	G-20M	- / 33.5	- / 28.0	- / 34.3	- / 33.9
T-Rex2-T [15]	Swin-T	OG	- / 34.9	- / 32.7	- / 32.9	- / 37.1
YWorldv2-S [3]	YOLOv8-S	OG	- / 24.4	- / 17.1	- / 22.5	- / 27.3
YWorldv2-M [3]	YOLOv8-M	OG	- / 32.4	- / 28.4	- / 29.6	- / 35.5
YWorldv2-L [3]	YOLOv8-L	OG	- / 35.5	- / 25.6	- / 34.6	- / 38.1
YOLOEv8-S [34]	YOLOv8-S	OG	25.7 / 27.9	19.0 / 22.3	25.9 / 27.8	26.7 / 29.0
YOLOEv8-M [34]	YOLOv8-M	OG	29.9 / 32.6	23.6 / 26.9	29.2 / 31.9	31.7 / 34.4
YOLOEv8-L [34]	YOLOv8-L	OG	33.3 / 35.9	30.8 / 33.2	32.2 / 34.8	34.6 / 37.3
YOLOEv11-S [34]	YOLOv8-S	OG	25.2 / 27.5	19.3 / 21.4	24.4 / 26.8	26.7 / 29.3
YOLOEv11-M [34]	YOLOv8-M	OG	30.5 / 33.0	22.4 / 26.9	30.4 / 32.5	32.1 / 34.5
YOLOEv11-L [34]	YOLOv8-L	OG	32.4 / 35.2	25.6 / 29.1	31.9 / 35.0	34.1 / 36.5
OV-DEIM-S [36]	ViT-T	OG	27.7 / 29.6	23.6 / 25.2	28.1 / 30.2	28.0 / 30.0
OV-DEIM-M [36]	ViT-T+	OG	30.6 / 32.6	25.3 / 26.9	30.2 / 31.5	31.9 / 34.1
OV-DEIM-L [36]	ViT-S	OG	33.7 / 35.9	34.3 / 36.8	33.4 / 35.5	34.0 / 36.0
VL-DINO-T (ours)	Swin-T	OG	36.3 / 38.7	37.8 / 40.6	35.6 / 37.9	36.7 / 39.0
VL-DINO-L (ours)	Swin-L	OG	38.1 / 40.2	39.2 / 41.8	38.1 / 40.9	37.7 / 40.1

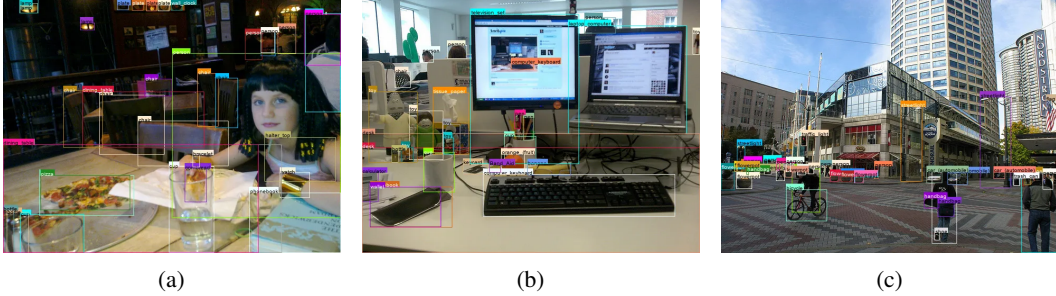


Figure 3: Visualizations of zero-shot inference on LVIS. The pretrained VL-DINO-T model is evaluated with all 1,203 category text embeddings as input.

frozen throughout training. Unless otherwise specified, we train VL-DINO on eight NVIDIA L40 GPUs using Objects365v1 [30] and GoldG [16] consistent with previous methods. For simplicity, we refer to the two datasets as O and G, respectively.

Evaluation. To assess the open-vocabulary detection performance of the pre-trained VL-DINO, we perform zero-shot evaluation on the COCO [20] val2017 split and the LVIS [9] minival split. These two benchmarks exhibit distinct class distributions and evaluation characteristics, and thus provide a comprehensive test bed for assessing the generalization ability of open-vocabulary detectors. For COCO, which contains 80 categories, we report the standard AP metric. For LVIS, which contains 1,203 categories, we report both standard AP and Fixed AP [4], with the maximum number of predictions per image set to 300 and 900, respectively.

4.2 Zero-shot results

LVIS. As shown in Table 1, VL-DINO achieves the best performance on LVIS in the zero-shot setting. VL-DINO-T obtains 36.3 AP and 38.7 Fixed AP, while VL-DINO-L further reaches 38.1 AP and 40.2 Fixed AP, surpassing all advanced models under this setting. Compared with GLIP-T and GDINO-T, VL-DINO-T achieves substantially better performance without resorting to their computationally intensive layer-wise cross-modal fusion modules or large-scale training corpora. At inference time, our method retains only extra VSE, yet still surpasses GLIP-T and GDINO-T by 12.7 and 11.3 Fixed AP, respectively. VL-DINO-T also outperforms recent YOLO-style detectors(e.g.,

Table 2: Zero-shot object detection performance comparison on COCO.

Model	Backbone	Data	AP
GLIP-T [19]	Swin-T	OG	46.7
GDINO-T [22]	Swin-T	OG	48.1
GDINO-L [22]	Swin-L	OG,OI	52.5
T-Rex2-T [15]	Swin-T	OG	46.4
YWorldv1-S [3]	YOLOv8-S	OG	37.6
YWorldv1-M [3]	YOLOv8-M	OG	42.8
YWorldv1-L [3]	YOLOv8-L	OG	44.4
OV-DEIM-S [36]	ViT-T	OG	40.8
OV-DEIM-M [36]	ViT-T+	OG	43.3
OV-DEIM-L [36]	ViT-S	OG	45.9
VL-DINO-T (ours)	Swin-T	OG	48.5
VL-DINO-L (ours)	Swin-L	OG	50.0

Table 3: Ablation study of QPSC, VSE, and ORSA, where modules are progressively introduced.

QPSC	VSE	ORSA	AP	AP _r	AP _c	AP _f
✗	✗	✗	31.8	26.3	31.2	33.4
✓	✗	✗	33.5	30.1	32.4	35.0
✓	✓	✗	35.6	35.1	35.8	35.6
✓	✓	✓	36.3	37.8	35.6	36.7

Table 4: Effect of QPSC on LVIS and COCO.

QPSC	Training Data	AP _{COCO}	AP _{LVIS}
✗	O	42.9	25.3
✓	O	44.9	25.7
✗	OG	44.5	31.8
✓	OG	45.4	33.5

YWorld and YOLOE) and DINOv3-style [31] detector(e.g., OV-DEIM), exceeding YOLOEv8-S by 10.8 Fixed AP and OV-DEIM-S by 9.1 Fixed AP, while consistently outperforming all reported variants. Moreover, on the challenging rare categories of LVIS, VL-DINO-L achieves the highest 41.8 Fixed AP, highlighting its strong effectiveness on long-tailed open-vocabulary detection. We present qualitative visualization results in Figure 3, demonstrating that VL-DINO-T can accurately localize multiple relevant objects across diverse scenes, while maintaining robust performance under varying object scales and complex visual contexts.

COCO. COCO mainly contains object categories with a relatively balanced class distribution, and thus serves as an important benchmark for evaluating zero-shot detection performance on frequent classes. As shown in Table 2, under the same OG training data, VL-DINO achieves strong performance on COCO in the zero-shot setting, with VL-DINO-T and VL-DINO-L reaching 48.5 and 50.0 AP, respectively. In particular, VL-DINO-T outperforms GLIP-T, GDINO-T, and T-Rex2-T under the same Swin-T backbone and OG training setup. Compared with recent YWorldv1 and OV-DEIM detectors, VL-DINO-T also shows clear advantages, surpassing YWorldv1-L by 4.1 AP and OV-DEIM-L by 2.6 AP. These results demonstrate that VL-DINO not only maintains strong open-vocabulary generalization, but also exhibits superior detection capability on frequent categories.

4.3 Ablation study

To better understand the contribution of each design in VL-DINO, we perform a series of ablation studies on QPSC, VSE, and ORSA. Unless stated otherwise, all ablation experiments are conducted using the VL-DINO-T trained on OG dataset and evaluated on LVIS in the zero-shot setting.

Overall component analysis. We first conduct a progressive component analysis on LVIS by incrementally introducing each proposed module into the baseline, so as to evaluate its individual contribution to the final performance. As shown in Table 3, incorporating QPSC and VSE yields gains of 1.7 AP and 2.1 AP, respectively. In particular, compared with the baseline, the AP_r score improves by 8.8 AP, suggesting that the richer vision-language supervision brought by QPSC and the visual semantic knowledge learned through VSE are especially effective for rare-category detection. It is also worth noting that, although ORSA improves the overall AP by only 0.7, it further boosts AP_r by 2.7 AP. This result indicates that the foreground object-region alignment in ORSA effectively encourages the model to learn more discriminative representations for rare categories.

Ablation study of QPSC under different training data settings. We analyze the effect of QPSC under different training data settings. As shown in Table 4, introducing QPSC consistently improves the zero-shot detection performance on both COCO and LVIS, regardless of whether the model is trained on O or OG. The gains are particularly noticeable on LVIS under the OG setting, where QPSC improves the AP from 31.8 to 33.5. These results verify that QPSC provides effective additional supervision and improves the model’s ability to learn transferable vision-language semantics.

Ablation study of VSE under different distillation losses. As shown in Table 5, introducing VSE alone, without distillation supervision, provides limited benefit. Specifically, applying VSE only to $\mathcal{P}_6$ achieves 33.7 AP, whereas extending it to both $\mathcal{P}_6$ and $\mathcal{P}_5$ slightly decreases the performance to

Table 5: Ablation study of VSE under different distillation loss settings and feature maps.

Distillation Loss	Feature Map	AP	AP _r	AP _c	AP _f
–	$\mathcal{P}_6$	33.7	30.0	32.5	35.2
–	$\mathcal{P}_6, \mathcal{P}_5$	33.3	29.4	32.0	34.8
$\mathcal{L}_{sem}, \mathcal{L}_{rel}$	$\mathcal{P}_6$	34.3	30.1	33.4	35.7
$\mathcal{L}_{sem}, \mathcal{L}_{rel}$	$\mathcal{P}_6, \mathcal{P}_5$	35.3	35.0	34.6	35.8
$\mathcal{L}_{sem}, \mathcal{L}_{rel}, \mathcal{L}_{spa}$	$\mathcal{P}_6$	34.7	34.1	33.8	35.8
$\mathcal{L}_{sem}, \mathcal{L}_{rel}, \mathcal{L}_{spa}$	$\mathcal{P}_6, \mathcal{P}_5$	35.6	35.1	35.8	35.6

Table 6: Ablation study of the number of self-attention blocks in VSE.

Block Num	AP	AP _r	AP _c	AP _f
1	34.7	33.8	33.9	35.5
2	35.3	34.8	34.4	36.1
3	35.6	35.1	35.8	35.6

Table 7: Ablation study of ORSA with different alignment losses.

Alignment Loss	AP	AP _r	AP _c	AP _f
$\mathcal{L}_{focal}$	36.0	35.6	35.7	36.4
$\mathcal{L}_{i2t}$	36.1	36.7	35.6	36.5
$\mathcal{L}_{i2t} + \mathcal{L}_{t2i}$	36.3	37.8	35.6	36.7

33.3 AP, with consistent drops in AP_r, AP_c, and AP_f. This suggests that VSE itself is insufficient to improve features, and may even introduce interference when no explicit semantic supervision is imposed. Once distillation losses are applied, however, the effectiveness of VSE becomes clear. Using $\mathcal{L}_{sem}$ and $\mathcal{L}_{rel}$ in Eq 2 already improves the performance, especially when both $\mathcal{P}_6$ and $\mathcal{P}_5$ are distilled jointly, where the AP rises to 35.3 and AP_r reaches 35.0, indicating that semantic and relational supervision effectively facilitate the transfer of CLIP visual knowledge. Further adding the spatial distillation loss $\mathcal{L}_{spa}$ in Eq 3 leads to the best overall result of 35.6 AP, while also achieving the highest AP_r of 35.1 and AP_c of 35.8. Overall, these results show that the gains of VSE mainly come from the proposed distillation objectives rather than vse structure itself, and that semantic, relational, and spatial supervision are complementary for learning richer visual semantic knowledge, especially on rare categories.

Ablation study of the number of self-attention blocks in VSE. Table 6 reports the effect of varying the number of self-attention blocks in VSE. As the number increases from 1 to 3, the overall AP improves steadily from 34.7 to 35.6. In particular, AP_r rises from 33.8 to 35.1, suggesting that a larger VSE is especially helpful for transferring visual semantics to rare-category detection. Although the 2-block setting achieves the highest AP_f of 36.1, the 3-block setting delivers the best overall AP together with the best AP_r and AP_c, indicating a better balance across different category groups. These results suggest that a shallow VSE is insufficient to fully model CLIP’s high-level visual semantics, while the 3-block design provides stronger semantic encoding capacity and is therefore adopted in our final model.

Ablation study of alignment loss in ORSA. Starting from the best VSE setting, which achieves 35.6 AP, 35.1 AP_r, 35.8 AP_c, and 35.6 AP_f, introducing ORSA with the initial focal loss [21] already brings further improvement to 36.0 AP and 35.6 AP_r, as shown in Table 7. This choice is reasonable because ORSA extracts holistic object-level foreground features, for which classification-style supervision can serve as a simple alignment objective. We then replace focal loss with contrastive alignment loss [24]. Although the overall AP changes only slightly, image-to-text loss improves AP_r to 36.7, and the bidirectional loss further boosts AP_r to 37.8 while achieving the best overall AP of 36.3. In comparison, the changes on AP_c and AP_f are relatively small, indicating that the benefit of contrastive supervision is much more pronounced on rare categories than on common or frequent ones. This suggests that ORSA mainly helps the model learn more discriminative semantic representations for challenging long-tail classes. Based on these results, we adopt the bidirectional contrastive loss as the default alignment loss in ORSA.

Ablation study of box scaling in ORSA. ORSA extracts object-region features using ground-truth box annotations. To examine whether introducing a moderate amount of surrounding context can improve vision-language alignment, we enlarge the width and height of the ground-truth boxes by different scale factors. As shown in Table 8, the default setting without enlargement achieves the

Table 8: Ablation of context-aware box scaling in ORSA.

Scale Factor	AP	AP _r	AP _c	AP _f
1.0×	36.3	37.8	35.6	36.7
1.1×	36.1	36.6	35.9	36.4
1.2×	35.8	36.1	34.9	36.5

Table 9: Ablation study of the number of foreground decoder layers in ORSA.

Layer Num	AP	AP _r	AP _c	AP _f
1	36.1	35.8	35.9	36.2
3	36.3	37.8	35.6	36.7
6	36.4	38.2	35.7	36.8

Table 10: Transfer learning results on COCO under both linear probing and full tuning settings.

Model	Epochs	AP	AP ₅₀	AP ₇₅
<i>Linear Probing</i>				
YOLOEv11-S [34]	10	37.0	52.9	40.4
YOLOEv11-M [34]	10	43.1	60.6	47.4
YOLOEv11-L [34]	10	45.1	62.8	49.5
VL-DINO-T (ours)	10	51.5	67.6	56.9
VL-DINO-L (ours)	10	53.1	69.8	58.8
<i>Full Tuning</i>				
YOLOEv11-S [34]	160	46.2	62.9	50.0
YOLOEv11-M [34]	80	51.3	68.3	56.0
YOLOEv11-L [34]	80	52.6	69.7	57.5
VL-DINO-T (ours)	15	55.9	72.1	61.7
VL-DINO-L (ours)	15	57.5	74.4	63.8

best overall performance, reaching 36.3 AP and 37.8 AP_r. Enlarging the box to 1.1× or 1.2× leads to consistent degradation in overall AP, with particularly clear drops on rare categories. Although the 1.1× setting slightly improves AP_c, its overall performance remains inferior. We conjecture that enlarging the box introduces additional background regions, causing the deformable encoder layer to encode background noise into the object-region features, which weakens visual feature representation. Therefore, we adopt the original ground-truth box without enlargement.

Ablation study of the number of foreground decoder layers in ORSA. Table 9 reports the effect of varying the layer number in ORSA. Compared with the baseline without ORSA, introducing only 1 layer already improves the overall AP from 35.6 to 36.1, demonstrating that foreground object-region alignment is effective for enhancing detection. Increasing the layer number to 3 further boosts the performance to 36.3 AP, with a particularly notable gain on rare categories, where AP_r reaches 37.8. This indicates that a deeper ORSA can provide stronger semantic refinement for challenging categories. Although extending ORSA to 6 layers yields the best numerical results, improving AP to 36.4 and AP_r to 38.2, the gain over the 3-layer setting is relatively small. Considering that a deeper ORSA also introduces more parameters and computation, we adopt 3 layers as the default setting, which offers a better trade-off between performance and complexity.

4.4 Transfer learning on COCO

Following YOLOE [34], we evaluate the transferability of pretrained models to COCO under both linear probing and full tuning settings, where linear probing only fine-tunes the classification layers while keeping the rest of the model frozen, and full tuning updates all model parameters. As shown in Table 10, VL-DINO consistently achieves the best performance in both regimes, demonstrating strong transferability of the pretrained representations. Under linear probing, VL-DINO-T and VL-DINO-L reach 51.5 AP and 53.1 AP, respectively, outperforming all compared baselines by clear margins. Under full tuning, VL-DINO further improves to 55.9 AP and 57.5 AP. In addition to the gains in AP, VL-DINO also achieves the best AP₅₀ and AP₇₅, indicating that the transferred representations benefit both classification and localization. Overall, these results indicate that VL-DINO learns highly transferable visual-semantic features that can be better adapted to downstream detection tasks.

5 Conclusion and limitation

We present VL-DINO, an open-vocabulary detector that enhances DINO by jointly leveraging CLIP’s textual and visual semantic knowledge. We introduce QPSC for richer textual supervision, VSE to distill CLIP visual features, and ORSA to align object-region representations with text. These designs effectively avoids over-reliance on a single modality and achieves stronger semantic transfer in open-vocabulary detection. VL-DINO offers an effective and scalable direction for integrating multimodal semantic priors into DETR-style open-vocabulary detectors. One limitation of our work is that the current framework does not incorporate more advanced visual foundation models, such as DINOv3 [31], to learn richer visual semantic knowledge. Since stronger visual foundation models may provide more discriminative and transferable representations, exploring their integration into our framework is left for future work.

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A Technical appendices and supplementary material

A.1 Mode implementation details

VL-DINO is trained on eight NVIDIA L40 GPUs. The total batch sizes for VL-DINO-T and VL-DINO-L are set to 64 and 32, respectively. For the visual detection branch, we use a pretrained Swin Transformer as the backbone, and encoder and decoder are composed of six deformable encoder layers and six deformable decoder layers, respectively, with 900 object queries used for object detection. The hyperparameters are shown in Table 11. We adopt CLIP-B/16 as the default foundation model for visual semantic distillation, while using its corresponding text encoder to extract textual features. To reduce the training cost, we pre-compute and cache the textual features offline, allowing the text encoder to be removed from the training pipeline. The classification confidence score is calculated based on the C -dimensional visual-text features as follows:

$$s = \sigma\left(\frac{\|\mathbf{v}\|_2 \|\mathbf{t}\|_2^\top}{\tau}\right) \quad (5)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, $\mathbf{v} \in \mathbb{R}^{900 \times C}$ denotes the object-query features output by the decoder, $\mathbf{t} \in \mathbb{R}^{M \times C}$ denotes M text features, and τ is a learnable parameter.

In QPSC, we generate auxiliary boxes using the same strategy as the denoising box generation mechanism in DINO. Since QPSC does not require negative samples, it uses only half the number of auxiliary queries compared with the denoising queries in DINO. We refer readers to DINO [40] for more implementation details of the box generation process.

In terms of the training objective, VL-DINO adopts focal loss as the classification loss $\mathcal{L}_{cls}$, $L1$ loss and GIoU loss as the box regression losses $\mathcal{L}_{box}$, $\mathcal{L}_{dis}$ in Eq. 4 as the distillation loss, and bidirectional contrastive loss as alignment loss $\mathcal{L}_{ORSA}$ in ORSA. Additionally, we apply auxiliary losses to both the output of encoder and each decoder layer. The overall loss is formulated as the sum of the above loss terms.

Table 11: Hyperparameters used in our experiments.

Item	Value
optimizer	AdamW
lr	1e−4
lr_image_backbone	1e−5
weight_decay	0.0001
clip_max_norm	0.1
encoder_layers	6
decoder_layers	6
dim_feedforward	2048
hidden_dim	256
dropout	0.0
nheads	8
number_queries	900
set_cost_class	1.0
set_cost_bbox	5.0
set_cost_giou	2.0
ce_loss_coef	2.0
bbox_loss_coef	5.0
giou_loss_coef	2.0
interm_loss_coef	1.0
τ_1	0.05
τ_2	0.1
num_label	100
λ_{sem}	1
λ_{rel}	1
λ_{spa}	2
λ_{ORSA}	1

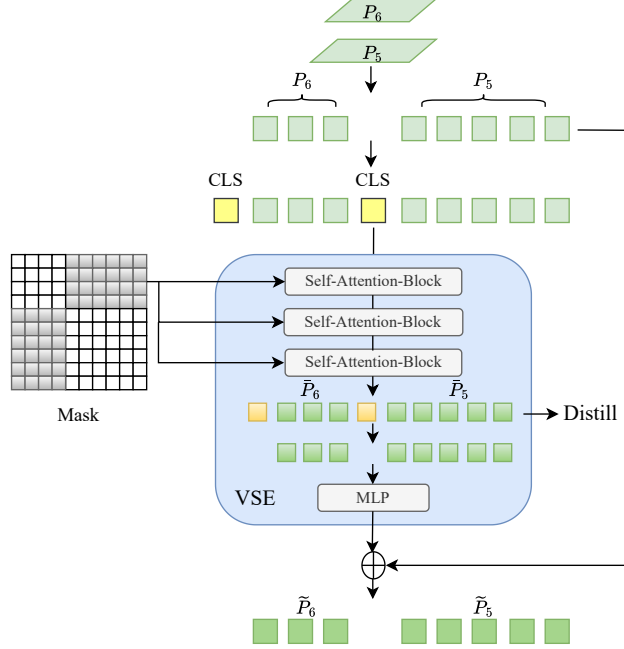


Figure 4: Details of the VSE. The attention mask prevents tokens from different feature levels from attending to each other, thereby avoiding cross-level interference between P_5 and P_6 . In the mask visualization, gray regions indicate blocked attention entries, while the remaining regions allow attention computation within the same feature level. The self-attention outputs are used for distillation, while the visual tokens are further passed through an MLP and fused with the original flattened tokens to produce $\tilde{P}_5$ and $\tilde{P}_6$.

A.2 VSE details

Figure 4 shows the detailed architecture of VSE. VSE is composed of three CLIP-style self-attention blocks followed by an MLP. Given the high-level features P_5 and P_6 , we first flatten them into token sequences and prepend a class embedding (CLS) to the tokens of each feature level. The resulting sequence is then processed by the self-attention blocks with an attention mask. The mask is designed to prevent tokens from different feature levels from attending to each other, so that the semantic modeling of P_5 and P_6 is performed independently without cross-level interference.

The outputs of the self-attention blocks are used for distillation. Afterwards, the CLS tokens are discarded, and the remaining visual tokens are passed through the MLP. The resulting tokens are fused with the original flattened tokens via residual addition, yielding the enhanced token features $\tilde{P}_5$ and $\tilde{P}_6$. This design allows VSE to enhance high-level semantic representations while preserving the original token organization and avoiding undesired interference between different feature levels.

A.3 ORSA details

As illustrated in Figure 5, the foreground decoder in ORSA adopts the same architecture as DINO decoder, while using only three decoder layers. Different from DINO decoder, it does not include box regression. Instead, ground-truth boxes are used as reference boxes to precisely extract foreground object-region features, which are then aligned with their corresponding textual features.

Note that, for clarity of illustration, Figure 5 depicts ORSA as being applied to the original image. During actual training, ORSA operates on feature maps to extract object-region features, instead of directly processing the original image.

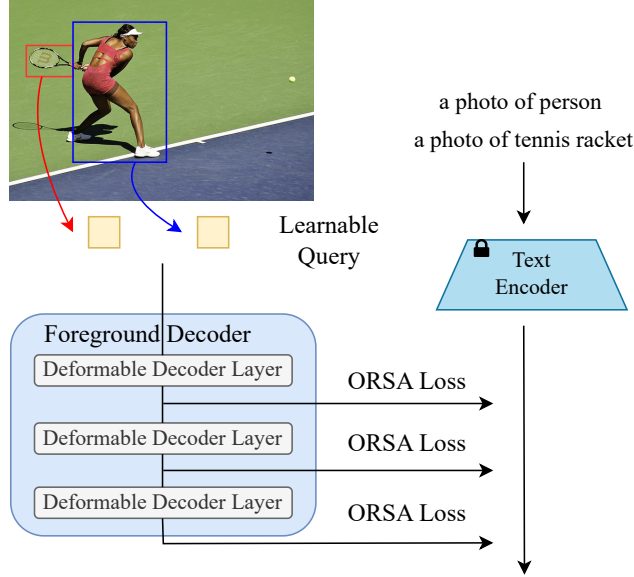


Figure 5: Framework of ORSA.

Table 12: Ablation study of different teacher models for visual semantic distillation in VL-DINO-T.

Teacher model	AP	AP_r	AP_c	AP_f
CLIP-B/16	36.3	37.8	35.6	36.7
CLIP-B/32	35.8	35.8	34.7	36.8
CLIP-L/14	36.6	37.0	36.4	36.6

A.4 Additional experimental results

A.4.1 Object query visualization

In Figure 6, we visualize the number of matches and average prediction scores of each query on the COCO and LVIS before and after introducing QPSC. The model is trained on the OG dataset until convergence. As shown in the figure, after introducing QPSC, object queries with smaller indices tend to produce higher prediction scores compared with the baseline without QPSC.

A.4.2 Ablation study of image teacher model

In addition to CLIP-B/16, we also use CLIP-B/32 and CLIP-L/14 as teacher models to distill visual semantic knowledge into VL-DINO-T. The experimental results are reported in Table 12. CLIP-B/32 brings a smaller performance improvement than CLIP-B/16. CLIP-L/14 further improves the results, but the gain remains limited. We conjecture that this is because the features extracted by the Swin Transformer differ from those of CLIP, and such a representational gap may restrict the amount of visual semantic knowledge that can be effectively transferred through distillation.

A.4.3 Inference

During both training and inference, VL-DINO can take pre-computed textual features cached offline as the classifier weights. Given an input image of size 640×640 and 1,203 textual features with a dimension of 512, we evaluate the inference performance of VL-DINO-T and VL-DINO-L on an NVIDIA L40 GPU, without including the text encoder. The corresponding performance statistics are shown in Table 13.

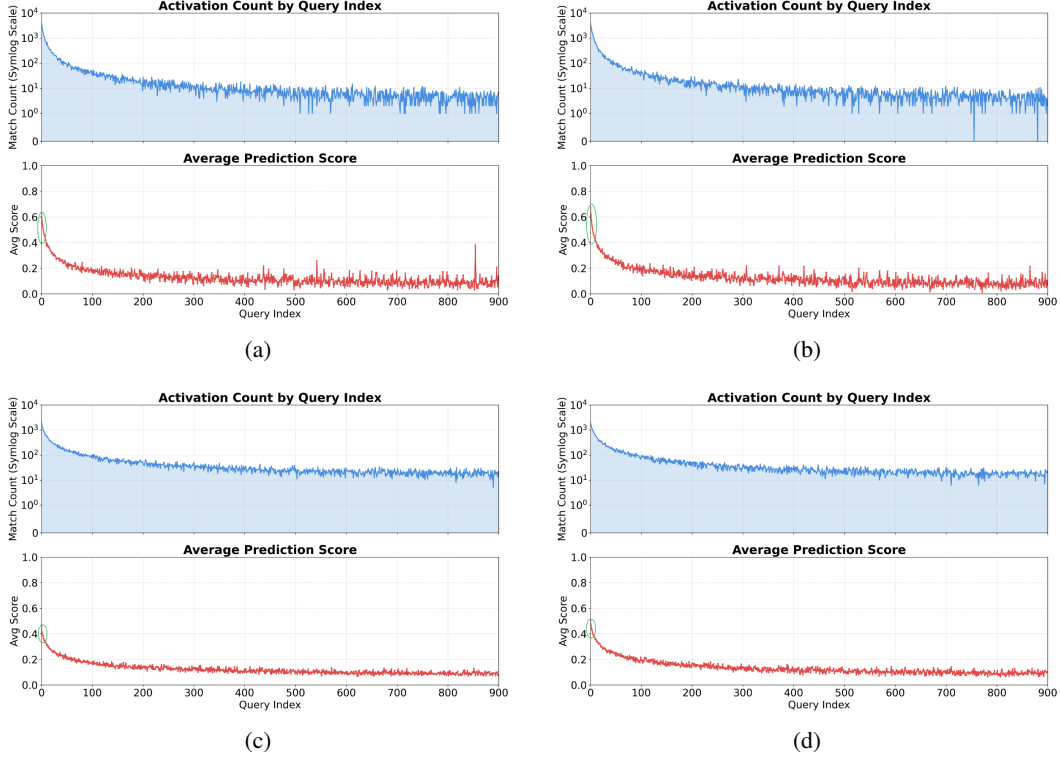


Figure 6: Object query statistics before and after introducing QPSC on COCO and LVIS. Subfigures (a)/(b) correspond to COCO without/with QPSC, while subfigures (c)/(d) correspond to LVIS without/with QPSC.

Table 13: Inference statistics of VL-DINO-T and VL-DINO-L.

Model	Params	GFLOPS	FPS
VL-DINO-T	51.4M	257G	27.8
VL-DINO-L	221.2M	811G	15.1

A.4.4 More visualization results

We provide additional detection visualization results on LVIS in Figure 7. During inference, the model takes an image and 1,203 textual features as input.

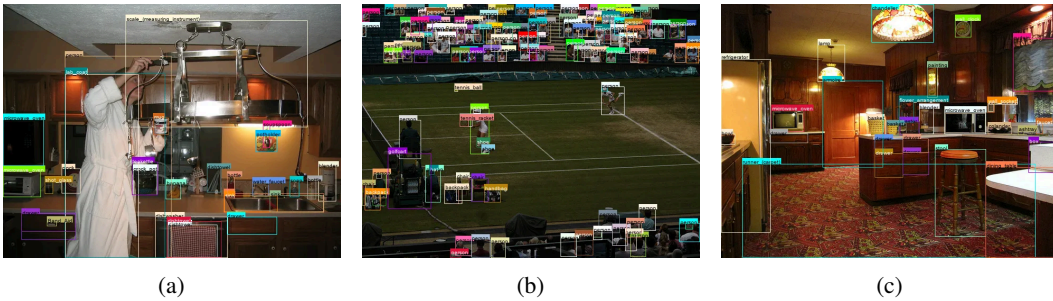


Figure 7: Visualizations of zero-shot inference on LVIS. The pretrained VL-DINO-T model is evaluated with all 1,203 category text embeddings as input.

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